

Raman Saparkhan

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EDUCATION

- **Carnegie Mellon University** Doha, Qatar
 - *Bachelor of Science in Computer Science, Dean's List* August 2022 – May 2026
 - **Relevant Coursework:** Distributed Systems, Deep Learning, Software Engineering, Machine Learning, Parallel & Sequential Data Structures and Algorithms, Linear Algebra, Multivariate Calculus, Great Theoretical Ideas in CS.
 - **Technical Leadership:** Google Developer Student Club Lead, Team Lead at AWS College Contributor Initiative.

EXPERIENCE

- **Qatar Computing Research Institution** Doha, Qatar
 - *Software Engineer Intern* December 2024 - Present
 - Built ML model integrating formal reasoning with **LLM & RAG** to boost logical consistency & reduce hallucinations.
 - Developed a hybrid CoT-PoT ensembling sampling approach that reduces LLM computational requirements by **95%**.
 - Enabled **multi-processing** & queuing to execute multiple models/datasets concurrently, boosting workflow efficiency.
- **Carnegie Mellon University, CS department** Doha, Qatar
 - *Teaching Assistant for Distributed Systems Course* August 2024 - November 2024
 - Assisted instructors in teaching material via office hours, 1:1 sessions, **introduction to Java**, and mentor meetings.
 - Applied big data frameworks **Ray, MapReduce, Hadoop, Google FS** to enhance distributed systems understanding.
 - Mentored 35 students throughout their course project, leading to special recognition from the instructors.
- **Carnegie Mellon University, Research Department** Doha, Qatar
 - *ML Research Engineer* May 2024 - July 2024
 - Enhanced Pairs Trading by applying Natural Language Processing into Quant analysis, achieving **15%** in annual return.
 - **Fine-tuned Llama 3-8B**, integrated financial LLMs & lexicons for sentiment analysis, boosting annual returns by 1.5%.
 - Built a data processing pipeline for 300 stock tickers using Refinitiv API, retrieving **300,000** news & time-series samples.
 - Developed custom **backtesting platform** for replicating algorithm design & optimization, enhancing work efficiency.
- **Amazon Web Services (AWS)** Seattle, WA, USA
 - *Software Engineer Contributor* March 2023 - May 2024
 - Contributed to OpenSearch **Python & Machine Learning** libraries enhancing operational efficiency & stability.
 - Developed support for **snapshot API**, redesigned test splits, integrated new python dependency management system.
 - Featured speaker at **OpenSearchCon 2023**, sharing insights into the open-source development for college contributors, increasing engagement.
- **Qatar Computing Research Institution** Doha, Qatar
 - *Software Engineer Intern* December 2022 - March 2023
 - Developed a cross-platform **Flutter** app for businesses to organize files using URLs, documents, tags & categories.
 - Optimized login page refresh & category retrieval logic, enhancing frame rendering time by **12%** on Flutter DevTools.
 - Migrated the app to null-safety and conducted end-to-end tests on **Android Studio**, utilized debugging tools, version control, and agile development with Cloud service **Azure DevOps**, increasing crash-free session rates by **20%**.

PROJECTS

- **Sketch AI** | [Demo Link](#) | *React, Node.js, Express, CSS, Stability AI, JavaScript, TypeScript*
 - Built a full-stack app with computer vision model that converts sketches and prompts into images for creative design.
 - Engineered a scalable Node.js & Express backend and **React/TypeScript** frontend to boost real-time image rendering.
- **Speech Recognition MLP** | *Python, PyTorch, Kaggle, Data Preprocessing, Neural Network Design, Linux*
 - Designed Multilayer Perceptron (MLP) to classify phonemes on Mel spectrograms data, achieving **86.6%** accuracy.
 - Built pipeline for speech preprocessing, including zero-padding, context windowing, improving phoneme recognition.
 - Enhanced model performance through hyperparameter tuning of architecture, activations, and optimization techniques.
 - Trained **Neural Networks** applying distributed computing on **NVIDIA A100 GPUs** on Linux Ubuntu Servers.
- **FileStack - Distributed File System** | *Java, Java RMI, TCP/IP, dynamic proxy*
 - Engineered Distributed File System (DFS) using **Java RMI & RPC** to address storage systems scalability challenges.
 - Programmed concurrency, synchronization, multi-threading, & object-oriented design, boosting system reliability & server performance by 30%.
- **SQL Query Generator** | [Demo Link](#) | *React, Node.js, Express, CSS, OpenAI API, JavaScript*
 - Built a tool to convert natural language into **SQL** queries, simplifying database interactions for non-technical users.
 - Designed a modular backend with **Node.js & Express**, ensuring seamless integration with responsive React interface.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, Dart, Rust, JavaScript, HTML, CSS, SQL, SML
- **Tools:** Git, PyTorch, TensorFlow, Numpy, Scikit-learn, Android Studio, HTTP, Pandas, OpenCV, Selenium
- **Technologies:** Node.js, Express, React, Flutter, Azure, AWS, Vercel, Langchain, Docker, Redis, Linux, Firebase